

Explore
Learn
Build
Create

IR SENSOR

A Tiny Eye That Sees Invisible Light!



Small
Sensor
Big
Possibilities!

What is an IR Sensor?

An IR sensor (Infrared Sensor) can detect invisible infrared light. It has an IR LED that sends out infrared light and a receiver that detects the reflected light back.

It works a bit like your eyes, but it sees invisible light!



Parts of an IR Sensor

IR LED
(sends out
infrared light)

IR Receiver
(detects
reflected light)

Potentiometer
(adjusts sensitivity)

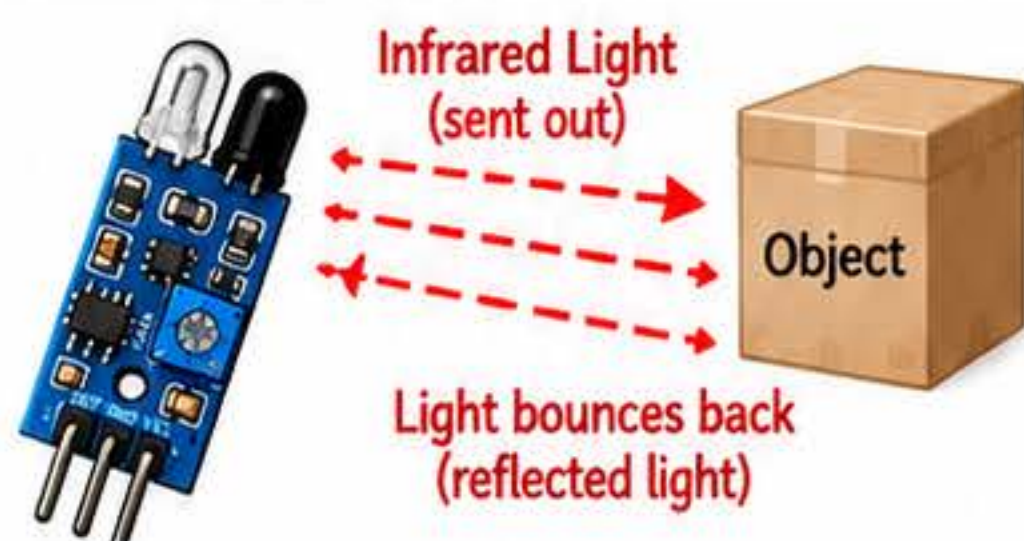
Indicator LED
(lights up when
it detects something)

OUT

GND

VCC

How Does It Work?



The IR LED sends out infrared light. When the light hits an object, it bounces back. The receiver detects the reflected light and the sensor gives a signal.

The Potentiometer

Turn
Left
(anti-clockwise)
→ less
sensitive



Turn
Right
(clockwise)
→ **more
sensitive**

The potentiometer lets you adjust how sensitive the sensor is. This means you can change how close an object needs to be before the sensor detects it.

What Can It Do?

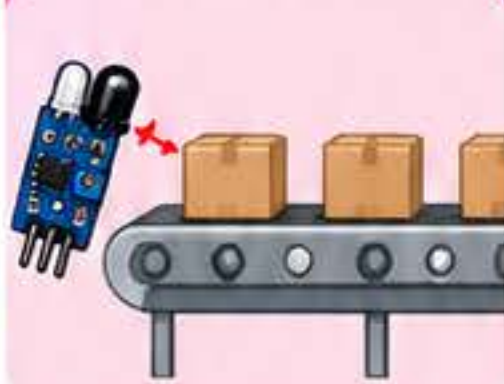
An IR sensor can be used to:

Detect Objects



It can tell when something is near.

Count Things



It can count objects like toys or boxes.

Find a Line



It can help robots follow a black line.

Make Things Automatic



It can be used in automatic doors, hand sanitizers and more!

Real-Life Examples



Automatic
Hand Sanitizers



Elevator
Doors



Line Following
Robots



Object Detection
Toys



Parking
Sensors

Fun STEM Fact!



Infrared light is invisible to our eyes, but many devices can see it! TV remotes also use infrared light just like an IR sensor.

Imagine • Experiment • Create a Smarter World!